

Multiple Choice (10 marks)

- Which of the following is lower for argon than for neon?
 - Melting point
 - Boiling point
 - Polarizability
 - First ionization energy
- When the following equation is balanced, which of the following is true? $\text{MnO}_4^- + \text{I}^- + \text{H}^+ \rightleftharpoons \text{Mn}^{2+} + \text{IO}_3^- + \text{H}_2\text{O}$
 - The $\text{I}^- : \text{IO}_3^-$ ratio is 3:1.
 - The $\text{MnO}_4^- : \text{I}^-$ ratio is 6:5.
 - The $\text{MnO}_4^- : \text{Mn}^{2+}$ ratio is 3:1.
 - The $\text{H}^+ : \text{I}^-$ ratio is 2:1.
- Calculate the Larmor frequency for a proton in a magnetic field of 1 T.
 - 23.56 GHz
 - 42.58 MHz
 - 74.34 kHz
 - 13.93 MHz
- Of the following compounds, which has the lowest melting point?
 - HCl
 - AgCl
 - CaCl_2
 - CCl_4
- Of the following compounds, which is LEAST likely to behave as a Lewis acid?
 - BeCl_2
 - MgCl_2
 - ZnCl_2
 - SnCl_2

True / False (10 marks)

Answer True or False

6. True or False: Tetramethylsilane is commonly used as a ^1H reference because its protons are weakly shielded in NMR spectra.
7. True or False: The hybrid sp^3 orbitals for carbon are formed from three atomic orbitals.
8. True or False: The ^{13}C NMR spectrum of 2-methylpentane exhibits five distinct chemical shifts due to its unique structural isomers.
9. True or False: Protons in a molecule with a rotational correlation time of 1 ns have the fastest spin-lattice relaxation at a magnetic field of 3.74 T.
10. True or False: A 0.100 M HCl solution has a higher ionic strength than a 0.050 M CaCl_2 solution because it is a strong acid.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors that contribute to the electron affinity of elements and analyze why calcium exhibits the lowest electron affinity among its peers.
12. Discuss the concept of conjugate acid-base pairs, providing an example, and analyze the significance of the pair H_3O^+ and H_2O in acid-base chemistry.

End of Paper